Project Proposal: Food Wastage Predictor

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Background

Food waste has become a serious and escalating issue in Malaysia, posing significant economic, environmental, and social challenges. According to the Solid Waste and Public Cleansing Management Corporation (SWCorp), Malaysia generates approximately 16,000–17,000 tonnes of food waste every day, accounting for nearly 44% of total municipal solid waste. A large portion of this waste originates from food service establishments such as cafeterias, restaurants, and institutional dining facilities, where food is prepared in bulk based on demand estimates rather than actual consumption patterns.

This problem is particularly evident in university cafeterias, including those at Universiti Teknologi PETRONAS (UTP). UTP cafeterias serve a large and fluctuating population of students, lecturers, and staff whose dining behaviour varies daily depending on class schedules, examination periods, weather conditions, academic calendars, and special campus events. As a result, cafeteria operators often overprepare meals to avoid shortages, leading to significant amounts of uneaten food, plate waste, and expired raw ingredients at the end of each day.

The consequences of this inefficiency are severe. From an economic perspective, food waste directly increases operational costs for cafeteria operators through unnecessary spending on raw materials, utilities, labour, and waste disposal. For institutions like UTP, these costs accumulate over time and ultimately affect food pricing and budget efficiency. From an environmental standpoint, wasted food represents wasted resources — including water, energy, and land used during food production — and contributes to landfill methane emissions, a greenhouse gas substantially more harmful than carbon dioxide. From a social perspective, the continued disposal of edible food highlights a critical imbalance, especially when food insecurity remains an ongoing concern among segments of the Malaysian population.

Despite the scale of the problem, most UTP cafeterias still rely on manual estimation and past experience to determine daily food preparation quantities. This raises critical questions:

- Why are cafeterias still unable to accurately predict daily food demand?
- Who bears the financial and environmental cost of repeated overproduction?
- How much food — and money — could be saved if demand were forecast more precisely?

The lack of a data-driven decision support system means cafeteria operators have limited visibility into consumption trends, high-waste menu items, and demand fluctuations. Without predictive tools, food waste continues to be treated as an unavoidable operational issue rather than a preventable and measurable problem.

Therefore, this project proposes the development of a Food Wastage Predictor system tailored for UTP cafeterias, leveraging data analytics and machine learning to forecast food demand more accurately. By aligning food preparation with actual consumption patterns, the system aims to reduce waste, lower operational costs, and support UTP's broader sustainability initiatives.

Objectives

The objectives of this project are to address the persistent problem of food overproduction and wastage in UTP cafeterias, which is primarily caused by inaccurate demand estimation and continued reliance on manual forecasting methods. This issue directly affects the efficiency, cost structure, and sustainability performance of cafeteria operations.

As a result, the **primary customers** of the proposed system are **UTP cafeteria operators and cafeteria management**, who are responsible for planning food preparation and managing operational resources. The **end users** include **cafeteria staff involved in daily food preparation**, inventory control, and purchasing decisions, who require clear and practical guidance to support their tasks. In addition, **UTP management and the wider campus community** are **indirect beneficiaries**, as reduced food wastage contributes to **lower operational costs, improved sustainability outcomes**, and **more responsible resource use** on campus.

The following objectives are defined to guide the development of the proposed solution and address the problem of food overproduction and wastage in UTP cafeterias:

1. To **develop a data-driven food demand prediction system for UTP cafeterias** that enables cafeteria operators to accurately estimate daily food preparation quantities using historical sales data, attendance patterns, and contextual factors such as academic schedules and weather conditions.

2. To **reduce food wastage in UTP cafeterias** by at least 20% within three months of implementation by improving alignment between food preparation levels and actual consumption demand.

3. **To enhance operational decision-making for cafeteria operators** through a user-friendly dashboard that presents demand trends, high-waste menu items, and inventory usage insights, supporting more efficient food preparation and purchasing decisions.

Scope

The project will result in a web-based food waste prediction system for cafeterias. Key phases include:

1. Data Collection: Gathering historical data on meal consumption, inventory, menu items, and student/staff attendance.
2. Data Analysis & Modeling: Cleaning data and developing a predictive model using machine learning techniques to estimate demand per menu item.
3. Dashboard Development: Creating a user-friendly interface to visualize predictions, track waste, and provide recommendations.
4. Testing & Evaluation: Implementing a pilot program in a selected cafeteria to test model accuracy and measure waste reduction.

Solution

1) **Predictive Alerts for Inventory Management**: Live dashboards allow clients/restaurants to adapt to the need to buy raw ingredients excessively or too low leading to less loss of capital due to understocking or overstocking

2) **Adaptive Statistics**: With the latest sales data updated each week by the client, the ML model will be refreshed, resulting in a new threshold for each raw item and serving prediction for each day.

3) **Environmental & Contextual Awareness**: The system will incorporate external and contextual factors that directly influence human behavior and demand patterns, such as weather conditions, academic schedules and special events.

4) Shelf-Life Prediction: Instead of treating all items as either *usable* or *expired*, the system applies Quality Decay Score, representing the gradual loss of functional or culinary value over time. System can suggest prioritizing materials nearing its shelf-life period.

5) **Dashboard for Visualization and Decision Support**:
The proposed dashboard delivers real-time visual insights through a simple and intuitive interface, allowing even entry-level staff to handle essential restaurant logistics efficiently. Clear indicators, minimal interaction steps, and fast access to key information enable staff to make timely decisions regarding food preparation and inventory management without requiring technical expertise.

Monitoring and Evaluation

FORMATIVE EVALUATION

1. Model Accuracy Checks (Weekly / Monthly)

- The predictive model will be evaluated regularly by comparing forecasted food demand against actual daily consumption data.
- Accuracy metrics such as percentage accuracy or error rate (e.g., MAE or MAPE) will be calculated.
- Weekly checks allow early detection of poor predictions, while monthly evaluations assess overall model stability.
- If accuracy drops below acceptable levels, the model will be retrained or adjusted using updated data.

Purpose:
To ensure the predictive model remains reliable and improves progressively during development.

2. Dashboard Usability Feedback from Cafeteria Staff

- Cafeteria staff will be asked to interact with the dashboard during testing phases.
- Feedback will be collected through:
  - Short questionnaires
  - Informal interviews
  - Observation of task completion (e.g., checking predictions, adjusting orders)
- Evaluation criteria include:
  - Ease of navigation
  - Clarity of predictions and alerts
  - Time required to make decisions

Purpose:
To confirm that the dashboard is understandable and usable by non-technical users, ensuring practical adoption.

3. Data Completeness and Update Frequency

- Incoming data (sales records, attendance, inventory) will be monitored to ensure:
  - No missing or inconsistent entries
  - Regular updates according to the planned schedule (e.g., daily or weekly)
- Data logs will be reviewed to identify gaps that may affect prediction accuracy.
- Any delays or inconsistencies will be documented and addressed.

Purpose:
To maintain high data quality, which is critical for accurate predictions and reliable system performance.

SUMMATIVE EVALUATION

1. Comparison of Predicted vs Actual Demand

- Final model performance will be evaluated by comparing:
    - Predicted daily food demand
    - Actual food consumption recorded by the cafeteria
- Results will be presented using:
    - Accuracy percentages
    - Error charts or tables
- Performance will be analysed across different menu items and days.

Success Indicator:
Prediction accuracy of at least 85%, as stated in the project objectives.

2. Measurement of Food Waste Reduction

- Food waste levels before system implementation will be compared with waste levels during the pilot period.
- Measurements may include:
    - Weight or quantity of uneaten food
    - Number of leftover portions per day
- The percentage reduction in waste will be calculated over the evaluation period.

Success Indicator:
A minimum 20% reduction in food waste within the pilot timeframe.

3. Staff Satisfaction and Dashboard Adoption

- Cafeteria staff will complete a post-project survey evaluating:
    - Overall satisfaction with the system
    - Perceived usefulness of predictions
    - Willingness to continue using the dashboard
- Adoption will be measured by:
    - Frequency of dashboard usage
    - Whether predictions are used to guide purchasing or meal preparation decisions

Success Indicator:
Positive usability and satisfaction feedback from the majority of staff, indicating successful adoption

Approval Signatures

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